

Pack a Cold-Chain Tote SOP

One episode packs one tote. The liner goes into the tote and lies flat on the floor, two gel packs cover the floor, the items go in on top of them, two more gel packs cover the items, the rest of the liner is folded over the top, the lid is laid on, the zipper is pulled shut, and the label is pressed onto the lid. The episode ends with a zipped, labelled tote in the middle of the tabletop, the staging area bare, and the gel spot bare.

The order never changes: liner, base gel, items, top gel, fold, lid, zip, label. Each layer goes in only after the one under it is down and complete. Nothing goes into the tote after the top gel is down, and the liner is never opened again once it is folded shut.

Each gripper has its own roles.

- The **left gripper** lifts the liner into the tote, loads every item, lays the lid on, and lays the label on.
- The **right gripper** lays all four gel packs, the two **base** gel packs and the two **top** gel packs, and presses the liner flat if it stops short of the floor.
- **The liner is folded flap by flap.** The part of the liner standing above the load has a far flap and a near flap. The far flap is folded in first, then the near flap, by one gripper or by both grippers together, until the two flaps meet over the load.
- **Zipping is a relay with one zipper pull.** The left gripper holds the tote while the right gripper runs the pull down the right side and stops at the near-right corner. Then they swap: the right gripper holds the tote while the left gripper takes the same pull along the near edge and up the left side to the far-left corner.

The liner and the items are staged together in one staging area: on the left of the tote, at the front centre, or on its right, depending on the config the episode is set up in (see the configs under Steps). The four gel packs lie in one pile on the **gel spot** on the other side of the tote: on the right in Config L and Config M, on the left in Config R. The label always lies at the **front centre**, in front of the tote. The gripper on the staging side takes the liner and the items, the gripper on the gel side takes the gel packs and presses the liner, and the left gripper takes the label and the lid in every config. In Config R the two grippers swap the liner, item, and gel roles listed above.

The tote stays on its spot in the middle for the whole episode, and a gripper only ever opens over the tote.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the tote in the middle and the whole staging area.
3. The open top of the tote is fully in frame from above, so the whole floor is visible and every piece can be seen going in.
4. Both arms are at home with their grippers open.
5. The tabletop is clear of everything except the zones listed below.
6. The **staging gripper** for this episode's config can reach the **far side of the tote floor**, the **liner spot**, and the **far end of the item row** without the arm leaning out over the tote. If it cannot, move the tote or the staging area until it can.
7. The **gel gripper** can reach the **far side of the tote floor**, the **middle of the tote floor**, and the **gel pile** the same way, and the **left gripper** can reach the **label spot** at the front centre.
8. The **right gripper** can reach the **zipper pull** at every point from the far-right corner to the near-right corner, and the **left gripper** can reach it at every point from the near-right corner along the near edge and up the left side to the far-left corner, without either arm going to the end of its reach, while the other gripper is holding the tote at the near corner on its own side.
9. The two arms clear each other over the tote. Test this with both grippers pinching the top edge of the liner's far flap, one at each end, both over the opening.
10. Nothing blocks the paths between the staging area and the tote.

Materials checklist

1. One **insulated tote** sits square on the tote spot in the middle of the tabletop, empty, with its **lid** unzipped and laid back over the far side, out of the opening. The **zipper pull** is parked at the far-right corner.
2. One **folded liner** lies on the liner spot in the staging area, closest to the tote, folded to about the size of the tote floor, with its opening edge toward the tote. The fold opens without tearing.
3. Four **gel packs**, each hard, flat, unpunctured, and dry on the outside, lie in one flat pile on the gel spot, on the other side of the tote from the staging area. Any pack can go in any layer. The pack on top of the pile is always the next one taken.
4. Each gel pack is about half the size of the tote floor, so two packs lying on it cover the floor to all sides.

5. The **items**, sealed and undamaged, lie in one row on the item spots in the staging area, in the order they are loaded, closest to the tote first.
6. One self-adhesive **label** lies flat on the label spot at the front centre, between the tote and the front edge, ready to pick up. It sticks when it is pressed onto the lid.
7. Before the session, confirm that the items lie on the gel without overlapping each other, and that the whole staged row plus both gel layers leaves the top of the load **below the tote rim** with room for the folded liner and the lid.

Workspace layout

- **Tote spot:** in the middle of the tabletop, centered between the two arms. The tote sits here for the whole episode with its lid laid back over the far side.
- **Staging area:** on the staging side for the episode's config: on the left of the tote in Config L, at the front centre in Config M, on the right of the tote in Config R.
- **Gel spot:** on the other side of the tote from the staging area: right of the tote in Config L and Config M, left of the tote in Config R. The four gel packs lie here in one flat pile, taken from the top down.
- **Label spot:** at the front centre, between the tote and the front edge, the same in every config.

Vocabulary

- **Arm assignment:** in Config L and Config M, the left gripper lifts the liner in and loads every item, and the right gripper lays the two base gel packs and the two top gel packs and presses the liner flat if it stops short of the floor. In Config R the two swap: the right gripper lifts the liner in and loads the items, and the left gripper lays the gel and presses the liner. In every config, the left gripper lays the lid on and lays the label on, the liner is folded far flap first, then near flap, by one gripper or both, and when zipping the left gripper holds the tote while the right gripper runs the pull to the near-right corner, then the right gripper holds while the left gripper runs the same pull the rest of the way. Both grippers press the label down. Neither gripper takes over the other's role.
- **Staging side:** the side of the tote the staging area is on: left in Config L, front centre in Config M, right in Config R.
- **Staging gripper:** the gripper on the staging side. It takes the liner and the items off the staging area: the left gripper in Config L and Config M, the right gripper in Config R.
- **Gel gripper:** the gripper on the same side as the gel spot. It lays all four gel packs: the right gripper in Config L and Config M, the left gripper in Config R.
- **Free gripper:** the gripper that is not carrying the liner in Step 1. It is the gel gripper: the right gripper in Config L and Config M, the left gripper in Config R. It presses the liner flat if the liner catches on the rim or stops short of the floor.

- **Reach rule:** the staging area is reached only by the staging gripper, the gel spot only by the gel gripper, and the label spot only by the left gripper. No arm reaches across the table for a staged piece. Both arms reach the inside of the tote, the liner when folding, and the lid. The right arm reaches the zipper pull from the far-right corner to the near-right corner, and the left arm reaches it from the near-right corner onward.
- **Tote:** the insulated carrier that gets packed. It sits on the tote spot for the whole episode.
- **Lid:** the insulated flap joined to the tote along the far rim. It folds over the opening toward the front edge and zips shut around the other three sides.
- **Zipper pull:** the one tab that closes the zipper. It starts at the far-right corner and runs down the right side, along the near edge, and up the left side. The lid is shut when the pull reaches the far-left corner.
- **Hold the tote:** the closed gripper presses down on the lid at the near corner on its own side and stays there, so the tote does not lift, turn, or slide while the other gripper pulls the zipper. A hold never pinches and never slides.
- **Label:** the self-adhesive label that goes on the lid once the zipper is shut. It always lies at the front centre and is always taken by the left gripper, the same way in every config: handled flat, pinched at the middle of its edge nearest the tote, laid flat on the middle of the lid, and pressed down by both grippers.
- **Liner:** the foil-faced bag that goes inside the tote. Everything cold goes inside the liner, never between the liner and the tote wall.
- **Seated:** the liner lies flat against the tote floor, with no part of it bunched under the middle or tenting up off the floor.
- **Far flap and near flap:** the two parts of the liner that stand above the load once it is packed. The far flap is on the far side of the tote and the near flap is on the side toward the front edge. Each is folded in over the load by its top edge.
- **Gel pack:** one frozen pack. It is handled flat, pinched at the middle of one long edge, and always laid down flat.
- **Base gel:** the two gel packs laid on the liner floor. They are the first two packs taken off the top of the gel pile.
- **Top gel:** the two gel packs laid on the items. They are the last two packs taken off the gel pile.
- **Item:** one sealed chilled unit. It is handled flat, pinched at the middle of its edge nearest the tote, and always laid on its largest face.
- **Layer:** one row of pieces lying across the tote. The layers go in this order: the base gel, the items, and the top gel. Items are laid beside each other first, and on top of each other only when there is no room left beside.
- **Covered:** every part of what is underneath has a piece lying over it, with no bare gap showing at the tote wall. For gel packs, a pack that is tilted or rests partly on the other pack still counts as covering, as long as both packs are lying down.

- **Flat and level:** the piece lies on its largest face with all of it touching what is under it. No corner is lifted, no edge rides up on a neighbour, and it does not rock when the gripper leaves.
- **Below the rim:** the whole top of the load sits under the top edge of the tote, so the liner can fold shut over it and the lid can lie flat without being pushed up.
- **Lower in:** the gripper carries the piece down into the tote until it touches what is under it, and only then opens. A piece is never let go above the tote.
- **Settled:** the piece stays put for 2 seconds after the gripper lifts away, with nothing springing back, rocking, or sliding.
- **Release point:** the inside of the tote, or the lid when the label is laid on. A gripper never opens anywhere else, in particular never over the staging area or over the bare tabletop.

Steps

Steps 1 through 6 pack one tote. Every item is taken from its row closest to the tote first, every gel pack is taken off the top of the pile, and every piece is lowered until it touches down before the gripper opens.

The staging area is in one of three places, with the gel spot on the other side of the tote, and that is the only thing that changes between configs:

- **Config L:** the staging area is on the **left** of the tote and the gel spot on its **right**.
- **Config M:** the staging area is at the **front centre**, between the tote and the front edge, and the gel spot on its **right**.
- **Config R:** the staging area is on the **right** of the tote and the gel spot on its **left**.

What stays constant:

- The tote sits on the tote spot in the middle, lid laid back over the far side, zipper pull at the far-right corner.
- The staging area holds the same pieces in the same order: the folded liner closest to the tote, then the item row. The gel pile lies on the gel spot on the other side of the tote, and the label at the front centre.
- **Same-side rule:** the gripper on the staging side takes the liner and the items off the staging area, and the gripper on the gel side takes the gel packs off the pile. In Config M the staging area is at the front centre, so the left gripper takes the liner and the items and the right gripper takes the gel from the right. The left gripper takes the label in every config. No arm reaches across the table.
- The free gripper presses the liner flat if it stops short of the floor. Laying the lid, folding the liner, the zipper relay, and the label are done the same way in every config.

Only Step 1, the gel steps (2 and 4), and the item step (3) carry an IF line. Every other line is the same in all three configs.

Step 1: Line the tote

Goal: the liner is open inside the tote, seated flat on the floor, with its top standing open above the rim.

- **IF the staging area is on the left (Config L):** the **left gripper** is the staging gripper and the **right gripper** is the free gripper.
- **IF the staging area is at the front centre (Config M):** the **left gripper** is the staging gripper and the **right gripper** is the free gripper.
- **IF the staging area is on the right (Config R):** the **right gripper** is the staging gripper and the **left gripper** is the free gripper.

Then, in all configs:

- The **staging gripper** pinches the **folded liner** at the middle of its opening edge, the edge nearest the tote, and lifts it straight up off the liner spot.
- The **staging gripper** carries it over the tote and lowers it in, opening up, until it touches the floor. Open the gripper and lift clear.
- **IF the liner catches on the rim or stops short of the floor:** the **free gripper** closes and moves over the **middle of the tote**.
- The **free gripper** lowers straight down onto the liner in the middle of the floor and presses until the liner lies flat on the floor. Lift straight up and clear.
- Leave the top of the liner standing open above the rim. Nothing is folded over the rim.

Check: the liner is seated flat on the floor, it is open wide enough to see the floor through the opening, and no part of it is twisted or bunched. If the liner sits crumpled or twisted, the staging gripper lifts it straight out and lowers it in again.

Step 2: Cover the floor with the base gel

Goal: two gel packs lie down on the liner floor and cover it to all sides.

- **IF Config L or Config M:** the **right gripper** lays both base gel packs. **IF Config R:** the **left gripper** lays them.
- The **gel gripper** pinches the gel pack on top of the pile at the middle of its long edge nearest the tote, lifts it flat, and carries it flat to the tote.
- The **gel gripper** lowers it into the tote, inside the liner, until it touches the liner floor, then opens and lifts clear.
- The **gel gripper** takes the next pack off the top of the pile the same way and lowers it in beside the first one until it touches the liner floor. Open the gripper and lift clear.

Carry each pack flat the whole way. A pack tipped on edge in the air has to be turned back down inside the tote, which drags the liner up off the floor.

Check: both packs lie down on the liner floor and together cover the floor to all sides, with no bare gap at the tote wall. It does not matter if a pack is tilted, or if one pack rests partly on top of the other, as long as both packs are lying down and the floor is covered. If a pack stands on edge, or leaves the floor bare at a wall, lift it straight out, lower it in again, and let it settle.

Step 3: Load the items

Goal: the staged items lie inside the liner on the base gel, in their staged order, with the top of the load below the tote rim.

- **IF Config L or Config M:** the **left gripper** loads every item. **IF Config R:** the **right gripper** loads every item.
- The **staging gripper** pinches the item closest to the tote at the middle of its edge nearest the tote, lifts it flat, and lowers it onto the base gel until it touches down. Open the gripper and lift clear.
- The **staging gripper** takes the next item the same way and lowers it in beside the first one until it touches down. Open the gripper and lift clear.
- When there is no room left beside the items already down, the **staging gripper** lays the next item on top of them, the same way.
- Keep going, closest to the tote first, until the item row in the staging area is bare.

Take the items in their staged order. Reaching past a nearer item for one further out knocks the row over in the staging area.

Check: every item lies inside the liner on its largest face, on the base gel or on the items below, and the top of the load sits below the tote rim with room for the top gel. An item that rests against or leans on a tote wall or a side of the tote is not a fault and is not corrected. If an item stands on edge, lift it straight out and lower it in again. If the load rides above the rim, lift the top items out, reseal the items below them, and load them again.

Step 4: Cover the items with the top gel

Goal: two gel packs lie down on top of the items and cover them, still below the tote rim.

- **IF Config L or Config M:** the **right gripper** lays both top gel packs. **IF Config R:** the **left gripper** lays them.
- The **gel gripper** pinches the gel pack on top of the pile at the middle of its long edge nearest the tote, lifts it flat, and lowers it onto the items until it touches down. Open the gripper and lift clear.
- The **gel gripper** takes the last pack off the pile the same way and lowers it onto the items beside the first one until it touches down. Open the gripper and lift clear.

Check: both packs lie down on top of the items and together cover the load to all sides, and the top of the load is still below the tote rim. It does not matter if a pack is tilted, or if one pack rests partly on top of the other, as long as both packs are lying down and no item shows. If a pack stands on edge, or leaves an item bare, lift it straight up and lower it again. Once the top gel is down, nothing else goes into the tote.

Step 5: Fold the liner shut

Goal: the far flap and the near flap of the liner are folded flat over the top gel and meet over the load, so no gel pack or item shows.

- The liner standing above the load has a **far flap** and a **near flap**. Fold the far flap first, then the near flap. Either flap may be folded by one gripper pinching it at the middle of its top edge, or by both grippers pinching it together, one at each end of its top edge.
- Pinch the **far flap** at its top edge, lift it, and fold it in toward the front edge, over the top gel, until it lands flat over the load. Open the gripper and lift clear.
- Pinch the **near flap** at its top edge the same way and fold it in toward the far side, over the load, until its edge meets the edge of the far flap over the middle of the tote. Open the gripper and lift clear.
- If the two flaps do not meet, or folding one flap lifts the other, pinch that flap at its top edge again and lay it down until the two edges meet.

Check: the liner lies flat over the top gel with no gel pack or item showing, the far flap is down with the near flap folded in to meet it over the load, and nothing stands up above the rim. If a flap springs back up, pinch it at its top edge and fold it in again. Once the liner is folded shut it is not opened again.

Step 6: Close the lid and label

Goal: the lid lies flat over the opening, the zipper is pulled shut the whole way round, and the label is stuck flat on the lid.

6.1 Lay the lid on

- The **left gripper** takes the **lid** at the middle of its free edge, lifts it off the far side, and folds it toward the front edge, flat over the tote opening, until its edge sits on the near rim. Open the gripper and lift clear.

6.2 Zip the right side

- The **left gripper** closes and holds the tote at the **near-left corner** of the lid. It turns so that it does not block the view of the zipper run on the right side. It stays there until the right gripper has let go of the pull and lifted clear.
- The **right gripper** pinches the **zipper pull** at the far-right corner.

- The **right gripper** pulls it toward the front edge along the right side of the tote and stops at the **near-right corner**. If the pull slips out of the gripper or the zipper stops partway, pinch the pull again and keep pulling from where it stopped.
- The **right gripper** opens and lifts clear of the pull. Then the **left gripper** lifts clear.

6.3 Zip the near edge and the left side

- The **right gripper** closes and holds the tote at the **near-right corner** of the lid, on the part already zipped, beside the pull. It stays there until the left gripper has let go of the pull and lifted clear.
- The **left gripper** pinches the **zipper pull** where it stopped at the near-right corner.
- The **left gripper** pulls it along the near edge, round the near-left corner, and up the left side of the tote until it stops at the **far-left corner**. If the pull slips out of the gripper or the zipper stops partway, pinch the pull again and keep pulling from where it stopped.
- The **left gripper** opens and lifts clear. Then the **right gripper** lifts clear.

Hold first, pull second, let go of the pull before letting go of the tote. A pull with nothing holding the tote drags the tote along the tabletop instead of closing the zipper.

Check: the lid lies flat over the whole opening, square to the tote, the pull sits at the far-left corner, and no part of the liner is caught in the zipper. If the pull stops partway, back it up a little and pull again with the other gripper still holding. If the liner is caught, pull the zipper back past the catch, tuck the liner in under the lid with the gripper that is not holding the tote, and zip again.

6.4 Apply the label

- The **left gripper** pinches the **label** at the middle of its edge nearest the tote, lifts it off the label spot, and carries it flat over the tote.
- The **left gripper** lowers it onto the **middle of the lid** until it touches down, then opens and lifts clear.
- **Both grippers** close and press down on the label, the left gripper on its left half and the right gripper on its right half, so it sticks to the lid. Lift both straight up and clear.

Check: the label lies flat on the middle of the lid, stuck down at every edge with no corner lifting. If a corner or an edge lifts, press it down again with the gripper on that side.

Step 7: End the episode

- Confirm the tote sits square on its spot with its lid zipped shut, the zipper pull at the far-left corner, the label stuck flat on the lid, and the staging area and the gel spot bare.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Peel the label off the lid and discard it. Unzip the tote, run the zipper pull back to the far-right corner, and lay the lid back over the far side of the tote, out of the opening.
2. Unfold the liner, take out the top gel, the items, the base gel, and the liner, and set them aside.
3. Wipe the inside of the tote, the tote spot, the staging area, the gel spot, and the tabletop dry. Melt water under the tote makes it slide on the tabletop.
4. Put the empty tote back square on the tote spot, lid laid back over the far side, the zipper pull at the far-right corner.
5. Reload the staging area, on the staging side for the next episode's config, closest to the tote first: the liner folded flat to about the size of the tote floor with its opening edge toward the tote, then the row of items flat on their largest faces in the order they are loaded. Lay a fresh label flat on the label spot at the front centre.
6. Put four gel packs in one flat pile on the gel spot, on the other side of the tote from the staging area.
7. Check each gel pack is hard, flat, dry on the outside, and unpunctured. Swap any pack that has gone soft for a frozen one.
8. Check the staged items plus both gel layers still leave the top of the load below the rim.
9. Replace anything damaged: a torn or holed liner, a leaking gel pack, a crushed or unsealed item, a label that will not stick, or a zipper that no longer runs freely.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the staging side: the violations below were written for Config L and Config M (left gripper takes the liner and the items, right gripper takes the gel from the right). The arm-role

cues will be rewritten later to cover Config R; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Wrong order

- **Visible cue:** a gel pack or an item goes in before the liner lies flat on the floor, an item goes in before both base gel packs are down, the top gel goes on before the item row is bare, the liner is folded before both top gel packs are down, the lid goes on before the liner is folded, the zipper is pulled before the lid lies flat, or the label goes on before the zipper pull reaches the far-left corner.
- **SOP rule broken:** Steps 1 to 6: liner, base gel, items, top gel, fold, lid, zip, label, and each layer goes in only after the one under it is complete.
- **Coaching note:** liner, gel, items, gel, fold, lid, zip, label. Nothing jumps the queue.

Violation: Wrong gripper for the piece

- **Visible cue:** the right gripper lifts the liner, an item, the lid, or the label; the left gripper presses the liner to the floor or lifts a gel pack; the right gripper runs the zipper pull past the near-right corner; the left gripper starts the pull before the right gripper has let go of it; or a gripper pulls the zipper while the other gripper holds nothing.
- **SOP rule broken:** Steps 1 to 6: the left gripper lifts the liner, loads the items, lays the lid, and lays the label; the right gripper lays the base gel and the top gel and presses the liner if it stops short; either gripper folds the liner; the right gripper runs the zipper pull to the near-right corner and the left gripper runs it the rest of the way, each while the other holds the tote.
- **Coaching note:** each gripper has its own roles. Left lifts, loads, lids, and labels; right presses and lays the gel.

Violation: Liner put in wrong

- **Visible cue:** the left gripper drops the liner in from above the rim, lowers it in with the opening down or sideways, lands it between the tote wall and the tote spot, or leaves it twisted or crumpled without lifting it out and lowering it again.
- **SOP rule broken:** Step 1: pinch the liner at the middle of its opening edge, lower it in opening up until it touches the floor, and lift it out and lower it again if it sits crumpled or twisted.
- **Coaching note:** opening up, all the way down, then let go. If it lands wrong, start it again.

Violation: Liner not pressed to the floor

- **Visible cue:** the liner catches on the rim or stops short of the floor and the episode moves on with no press; the free gripper presses with its fingers open, presses at an edge or corner instead of the middle, or lifts away while the liner still tents off the floor; or a liner edge is folded down over the rim before loading.

- **SOP rule broken:** Step 1: if the liner catches on the rim or stops short of the floor, the closed free gripper presses it flat in the middle of the floor, and the top of the liner is left standing open above the rim.
- **Coaching note:** if it stops short: closed gripper, middle of the floor, straight down. Leave the top standing.

Violation: Something loaded outside the liner

- **Visible cue:** a gel pack or an item is lowered between the liner and the tote wall, or slides down the outside of the liner instead of into it.
- **SOP rule broken:** Steps 2, 3, and 4: everything cold goes inside the liner.
- **Coaching note:** look for the foil under the piece before you open the gripper.

Violation: Wrong number of gel packs in a layer

- **Visible cue:** one or three gel packs go on the floor or on top of the items, or a gel pack is left on the pile at the end.
- **SOP rule broken:** Steps 2 and 4: two base gel packs on the floor, then two top gel packs on the items, all four from the pile.
- **Coaching note:** two under, two over, all four from the pile.

Violation: Gel pack laid wrong

- **Visible cue:** a pack is left standing on edge or folded, a pack is carried tipped on edge through the air, or the layer is left with a bare gap at the tote wall or an item showing. A pack that is tilted, or rests partly on the other pack, is not a fault as long as both packs are lying down and the layer is covered.
- **SOP rule broken:** Steps 2 and 4: carry each pack flat and lay it down beside the other, so the two packs cover the layer to all sides.
- **Coaching note:** flat in the air, down on the layer, cover it all. Look across the whole layer before you move on.

Violation: Item loaded out of order or in the wrong place

- **Visible cue:** the items are taken out of their staged order, the arm reaches past a nearer item for one further out, or an item is laid on top of the items already down while there is still room beside them.
- **SOP rule broken:** Step 3: take the items closest to the tote first, lay each one beside the items already down, and go on top only when there is no room left beside.
- **Coaching note:** nearest first. Beside first, on top only when there is no room beside.

Violation: Item not laid flat

- **Visible cue:** an item is left standing on edge or upright inside the tote instead of lying on its largest face. An item that rests against or leans on a tote wall or a side of the tote is not a

fault.

- **SOP rule broken:** Step 3: each item lies on its largest face on what is under it.
- **Coaching note:** flat on its big face, every time. Touching the wall is fine; standing on edge is not.

Violation: Piece dropped in instead of lowered

- **Visible cue:** the gripper opens above the tote and a gel pack, an item, or the label falls the last part of the way, bounces, or lands askew.
- **SOP rule broken:** Steps 2, 3, 4, and 6: lower each piece until it touches down, then open the gripper.
- **Coaching note:** touch down first, then let go.

Violation: Load left proud of the rim

- **Visible cue:** the top of the load rides above the tote rim and the liner is folded shut over it anyway, or the lid is pushed down onto it to make the zipper close.
- **SOP rule broken:** Steps 3, 4, and 5: the top of the load sits below the rim before the liner is folded shut.
- **Coaching note:** if it stands above the rim, take the top items out and reseal them. Do not force the liner or the lid over it.

Violation: Liner not folded shut

- **Visible cue:** the lid goes on with a gel pack or an item still showing, a liner flap still standing above the rim, the two flaps left short of meeting, or the near flap folded in before the far flap; or a flap that springs back up is left standing.
- **SOP rule broken:** Step 5: fold the far flap in, then the near flap, flat over the top gel until they meet, so nothing shows and nothing stands above the rim.
- **Coaching note:** look down into the tote before the lid goes on. If you can see foil standing or gel showing, fold again.

Violation: Tote reopened after it was closed

- **Visible cue:** the folded liner is opened again, a gripper goes back inside the tote after the top gel is down, or anything is added, taken out, or rearranged after the liner is folded shut.
- **SOP rule broken:** Steps 4 and 5: nothing goes into the tote after the top gel is down, and the liner is not opened again once it is folded shut.
- **Coaching note:** the top gel closes the load. After the fold, the inside of the tote is finished.

Violation: Lid laid wrong

- **Visible cue:** the lid is left folded back, lands crooked or short of the near rim, is laid on with its edge over the liner instead of the rim, or is pressed down by the gripper to make it fit; or

the lid is taken anywhere other than the middle of its free edge.

- **SOP rule broken:** Step 6: take the lid at the middle of its free edge and fold it flat over the whole opening until its edge sits on the near rim.
- **Coaching note:** middle of the edge, over and down, flat on the rim.

Violation: Zipper pulled wrong

- **Visible cue:** the zipper is pulled before the lid lies flat, the pull is run while the other gripper is not holding the tote, the right gripper carries the pull past the near-right corner, the left gripper pulls before the right gripper has let go, the pull is pulled the wrong way, or it stops short of the far-left corner. Letting go of the pull and pinching it again partway is not a fault.
- **SOP rule broken:** Step 6: after the lid lies flat, the left gripper holds the tote while the right gripper runs the pull to the near-right corner, then the right gripper holds while the left gripper runs it to the far-left corner.
- **Coaching note:** lid flat first. Hold, then pull. Right gripper to the corner, left gripper the rest of the way.

Violation: Liner caught in the zipper

- **Visible cue:** a pull is dragged over a piece of liner sticking out of the opening and the episode moves on, or the liner is torn or pulled by the zipper instead of being tucked in.
- **SOP rule broken:** Step 6: no part of the liner is caught in the zipper; if it is, pull the zipper back past the catch, tuck the liner in, and zip again.
- **Coaching note:** stop as soon as the pull hits foil. Back up, tuck, zip again.

Violation: Label missing or not pressed on

- **Visible cue:** the episode ends with no label on the lid, the label is laid anywhere other than the middle of the lid, it is laid on and not pressed down by both grippers, or it is left with a corner or an edge lifting.
- **SOP rule broken:** Step 6: the left gripper lays the label flat on the middle of the lid, and both grippers press it down so it sticks.
- **Coaching note:** lay it flat, then press with both. A label that is not pressed does not stay.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the staging area, over the bare tabletop, or over open space, and a gel pack, an item, the liner, or the label lands anywhere other than inside the tote or on the lid.
- **SOP rule broken:** Steps 1 to 6: a gripper opens over the tote and nowhere else.
- **Coaching note:** know where it will land before you open the gripper.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the liner unfolded, the lid open, the zipper pull short of the far-left corner, no label on the lid, a piece still in the staging area, or an arm away from home.
- **SOP rule broken:** Step 7: confirm the zipped lid, the pull at the far-left corner, the label on the lid, and the bare staging area, return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table: the staging area or the gel spot is not on the side for the config, a gripper reaches across the table to take the liner, a gel pack, or an item, the wrong gripper presses the liner, or the wrong IF line is followed.
- **SOP rule broken:** the staging side and the same-side rule (the staging gripper takes the liner and the items, the gel gripper takes the gel packs, the free gripper presses the liner, no arm reaches across the table for a staged piece, and the IF line followed is the one for the config on the table).
- **Coaching note:** look where the staging area and the gel pile are before the first reach, then follow that config's IF lines through Steps 1, 2, 3, and 4.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, a controller-caused collision, or a motor error.
- **Defective liner:** torn, holed, or stuck shut so it will not open. Replace it before the next episode.
- **Defective gel pack:** soft, thawed, leaking, punctured, or frosted over so it will not lie flat. Replace it before the next episode.
- **Gel pack cut wrong for the tote:** too long to lie flat in the tote, or so wide that two packs cannot sit side by side on the tote floor.
- **Defective item:** crushed, split, unsealed, damp, or too tall to leave the load below the rim.
- **Staged load too big for the tote:** the staged items and both gel layers cannot fit below the rim.
- **Defective label:** torn, or one that will not stick to the lid when pressed. Replace it before the next episode.

- **Defective tote:** a zipper that jams or comes off its teeth, a torn lid, or a tote that will not sit square on the tote spot.

Annotation subtasks (from SOP)

1. Lift the liner into the tote
2. Press the liner flat to the tote floor
3. Lower one gel pack flat into the tote
4. Lower one item flat into the tote
5. Fold one liner flap in over the load
6. Lay the lid over the opening
7. Hold the tote and run the zipper pull along one gripper's share of the run
8. Lay the label on the lid and press it on
9. Return both arms home and end the episode